

Use of binary variables

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Lecture 8

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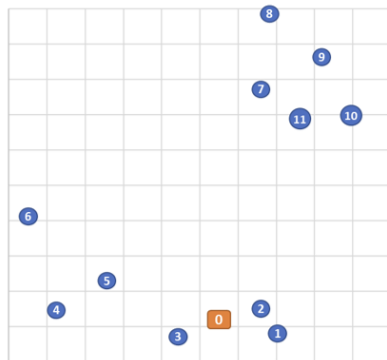
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1 TSP and Vehicle Routing Problem (VRP)

- Classic puzzles in mathematical programming
- What is the most efficient way for
 1. a vehicle (TSP)
 2. a fleet of vehicles (VRP)
 to deliver goods to a specific set of customers?
- We have to leave a central depot, hitting various stops (customers), and return home
- The goal is to minimize the total cost – usually measured by distance travelled, time spent, or the number of trucks used – while ensuring every customer gets served

1.1 TSP

- A company needs to plan the routes for a salesman visiting 11 customers, starting from the office and returning to the office
- Each customer can be visited only once
- Which route will minimize the total distance travelled?



Distance matrix:

	0	1	2	3	4	5	6	7	8	9	10	11
0	0	2	1	1	4	3	6	7	9	8	7	6
1	2	0	1	3	6	5	7	7	9	8	6	6
2	1	1	0	2	5	4	7	6	8	7	6	5
3	1	3	2	0	3	2	5	7	9	9	8	7
4	4	6	5	3	0	2	3	8	10	10	9	8
5	3	5	4	2	2	0	3	7	9	9	8	7
6	6	7	7	5	3	3	0	7	8	9	9	8
7	7	7	6	7	8	7	7	0	2	2	2	1
8	9	9	8	9	10	9	8	2	0	2	4	3
9	8	8	7	9	10	9	9	2	2	0	2	2
10	7	6	6	8	9	8	9	2	4	2	0	1
11	6	6	5	7	8	7	8	1	3	2	1	0

Sets

- Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a complete graph where:
- $\mathcal{V} = \{1, 2, \dots, n\}$: the set of n customers
- $\mathcal{E}$: the set of all edges (pairs of customers)

Parameters:

- $d_{i,j}$: distance from i to j

Variables:

- $x_{i,j}$: 1 if travelling from i to j , 0 otherwise
- u_i : the position of customer i in the tour sequence (auxiliary continuous variable)

Model

$$\begin{aligned} \min Z &= \sum_{i \in \mathcal{V}} \sum_{j \in \mathcal{V}, j \neq i} d_{i,j} x_{i,j} \\ &\text{s.t.} \\ &\sum_{j \in \mathcal{V}, j \neq i} x_{i,j} = 1 \quad \forall i \in \mathcal{V} \end{aligned} \tag{1}$$

$$\sum_{i \in \mathcal{V}, i \neq j} x_{i,j} = 1 \quad \forall j \in \mathcal{V} \tag{2}$$

$$u_i - u_j + nx_{i,j} \leq n - 1 \quad \forall i, j \in \{2, \dots, n\}, i \neq j \tag{3}$$

$$1 \leq u_j \leq n - 1 \quad \forall j \in \{2, \dots, n\} \tag{4}$$

1. Leave every customer exactly once – each node has exactly one outgoing edge
2. “Enter” every customer exactly once – each node has exactly one incoming edge

These constraints ensure a set of tours that covers all customers, but they don’t guarantee a single connected tour. Therefore we eliminate subtours with (3) and (4). This is the elegant trick – if $x_{i,j} = 1$ (i.e., we travel from i to j), then $u_j \geq u_i + 1$, enforcing a strict ordering of customers. A subtour would require a customer to appear “before himself” in the sequence – a contradiction. Customer 1 is fixed as the depot, so u_1 is excluded.

1.2 VRP

In VRP, there is a given number of “vehicles”. Each customer has a demand, and each vehicle may have limited capacity (in terms of total distance or in terms of cargo)

VRP can be modelled in different ways, and with different assumptions. Here is one example: **Sets and indices**

- $\mathcal{V} = \{1, \dots, |\mathcal{V}|\}$: set of vehicles
- $\mathcal{N} = \{0, 1, \dots, n\}$: set of nodes, where 0 is the depot and $\{1, \dots, n\}$ are customers

Parameters

1. $d_{i,j}$: distance from node i to node j
2. q_j : demand of customer j
3. Q : capacity of each vehicle

Decision variables

- $x_{v,i,j} \in \{0, 1\} \quad \forall v \in \mathcal{V}, i \in \{0, \dots, n\}, j \in \{0, \dots, n\}$: 1 if vehicle v travels from node i to node j , else 0

Model

$$\begin{aligned} \min Z &= \sum_{v \in \mathcal{V}} \sum_{i=0}^n \sum_{j=0}^n d_{i,j} x_{i,j} \\ &\text{s.t.} \end{aligned}$$

$$\sum_{i=0}^n \sum_{j=1}^n q_j x_{v,i,j} \leq Q \quad \forall v \in \mathcal{V} \quad (5)$$

$$\sum_{j=1}^n x_{v,0,j} = 1 \quad \forall v \in \mathcal{V} \quad (6)$$

$$\sum_{i=1}^n x_{v,i,0} = 1 \quad \forall v \in \mathcal{V} \quad (7)$$

$$\sum_{j=0}^n x_{v,i,j} \leq 1 \quad \forall v \in \mathcal{V}, i \in \{0, \dots, n\} \quad (8)$$

$$\sum_{i=0}^n x_{v,i,j} \leq 1 \quad \forall v \in \mathcal{V}, j \in \{0, \dots, n\} \quad (9)$$

$$\sum_{v \in \mathcal{V}} \sum_{i=0}^n x_{v,i,j} = 1 \quad \forall j \in \{1, \dots, n\} \quad (10)$$

$$\sum_{v \in \mathcal{V}} \sum_{j=0}^n x_{v,i,j} = 1 \quad \forall i \in \{1, \dots, n\} \quad (11)$$

$$\sum_{j=0}^n x_{v,i,j} = \sum_{j=0}^n x_{v,j,i} \quad \forall v \in \mathcal{V}, i \in \{0, \dots, n\} \quad (12)$$

$$x_{v,i,i} = 0 \quad \forall v \in \mathcal{V}, i \in \{0, \dots, n\} \quad (13)$$

5. Capacity
6. One departure from depot
7. One arrival to depot
8. At most outgoing arc
9. At most incoming arc
10. Each customer visited once (arriving)
11. Each customer visited once (departing)
12. Flow conservation
13. No self-loops

Subtour elimination (by subtour length)

$$\begin{aligned}
x_{v,i,j} + x_{v,j,i} &\leq 1 \quad \forall v \in \mathcal{V}, i, j \in \{1, \dots, n\} \\
x_{v,i,j} + x_{v,j,k} + x_{v,k,i} &\leq 1 \quad \forall v \in \mathcal{V}, i, j, k \in \{1, \dots, n\} \\
x_{v,i,j} + x_{v,j,k} + x_{v,k,l} + x_{v,l,i} &\leq 1 \quad \forall v \in \mathcal{V}, i, j, k, l \in \{1, \dots, n\} \\
x_{v,i,j} + x_{v,j,k} + x_{v,k,l} + x_{v,l,m} + x_{v,m,i} &\leq 1 \quad \forall v \in \mathcal{V}, i, j, k, l, m \in \{1, \dots, n\}
\end{aligned}$$

1.2.1 The “savings” heuristic, as suggested by Clarke and Wright

Assume the following cost matrix:

$C_{i,j}$	Depot	1	2	3	4
Depot		12	10	7	7
1	12		11	16	6
2	10	11		8	12
3	7	16	8		13
4	7	6	12	13	

and the following customer demands

Customer	Demand
1	10
2	8
3	5
4	16

Assume that the capacity of the truck is 32 pallets. First, combine according to the ranking list: 1 and 4: total cargo is 26 pallets.

Next on the list is combining 1 and 2, i.e., add customer 2 to the existing route if feasible. But not feasible since the remaining capacity of the truck (6 pallets) is lower than the demand (8 pallets) of customer 2.

Next on the list is a combination of 2 and 3. This will be a separate route since 2 and 3 are not on the first route. A route consisting of 2 and 3 is feasible since the total demand (13 pallets) does not exceed the truck's capacity.

All customers are now assigned to a route. Hence, the heuristic is finished.

The first route will then consist of (1) and (4). The total cost is $12 + 6 + 7 = 25$ (depot to 1 to 4 to depot). The second route will consist of (2) and (3). Total cost: $10 + 8 + 7 = 25$ (depot to 2 to 3 to depot). The total cost for this solution is $25 + 25 = 50$

Sets and Indices

- $\mathcal{N} = \{1, \dots, n\}$: customers
- $\mathcal{N}_0 = \{0\} \cup \mathcal{N}$: all nodes, depot indexes by 0
- $\mathcal{S} = \{(i, j) | i, j \in \mathcal{N}, i < j\}$: set of candidate savings pairs

Parameters

- $d_{i,j}$: distance between nodes i and j
- q_i : demand of customer $i \in \mathcal{N}$
- Q : vehicle capacity
- $s_{i,j} = d_{0,i} + d_{0,j} - d_{i,j}$: saving from merging i and j into one route

Decision Variable

- $y_{i,j} \in \{0, 1\} \quad \forall (i, j) \in \mathcal{S}$: 1 if the saving (i, j) is realized, i.e., customers i and j are placed on the same route with the depot-return between them eliminated

Objective function

$$\begin{aligned} \max Z &= \sum_{(i,j) \in \mathcal{S}} s_{i,j} y_{i,j} \\ \text{s.t.} \end{aligned}$$

$$\sum_{j \in \mathcal{N} \setminus \{i\}} y_{i,j} \leq 2 \quad \forall i \in \mathcal{N} \quad (14)$$

$$\sum_{j \in \mathcal{N} \setminus \{i\}} y_{i,j} \leq 1 \quad \forall i \in \mathcal{N} \quad (15)$$

$$\sum_{i \in mc} \quad (16)$$